K-Means Clustering Analysis with Initialization, Elbow, and LDA Projection

Exploring K-means clustering behaviour across initialization strategies, cluster counts, data distributions, and a discriminant-space projection on a real texture dataset.

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Overview

K-means is one of the most widely used unsupervised learning algorithms, yet its behaviour depends heavily on initialization, the choice of k, and the structure of the data. This project investigates those dependencies through five controlled experiments: clustering well-separated Gaussian blobs, varying the number of clusters, analysing the inertia elbow, testing on structureless uniform data, and applying K-means to the texture dataset (5500 samples, 40 features, 11 classes) both on raw features and after a linear discriminant analysis projection.

Goals

Demonstrate how initialization strategy, cluster count, and feature representation each affect K-means partition quality, and show that a supervised dimensionality-reduction step can turn a difficult clustering problem into a straightforward one.

Objectives

- Compare random vs k-means++ initialization on five well-separated 3-D Gaussian groups using the adjusted Rand index across repeated runs.
- Measure how the adjusted Rand index and inertia change as the number of clusters varies from 2 to 10, identifying the elbow and the peak ARI at the true k.
- Show that K-means on uniform (structureless) data produces unstable, meaningless partitions, contrasting with the Gaussian case.
- Apply K-means to the raw 40-feature texture dataset and record the baseline ARI against the 11 known classes.
- Project the texture data into a 10-dimensional LDA space and re-run K-means, measuring the improvement in ARI.

Approach

Each experiment is self-contained in a Jupyter notebook. Synthetic datasets are generated with scikit-learn make_blobs; the texture dataset is fetched from OpenML. The adjusted Rand index is used as the primary quality metric throughout, with inertia added for the elbow analysis. The LDA projection is fitted on the labelled training data and applied before K-means to maximise between-class separation.

Outcomes

- k-means++ initialization is consistently more stable than random on well-separated Gaussian groups, with ARI near 1 across runs.
- The inertia elbow and peak ARI both recover the true number of clusters ($k = 5$) without using labels.
- Uniform data yields highly variable ARI across runs, confirming that K-means partitions are only meaningful when genuine cluster structure exists.
- LDA projection improves ARI on the texture dataset from approximately 0.46 on raw features to approximately 0.99, nearly perfect recovery of the 11 texture classes.
- Scatter plots coloured by true class and by K-means cluster are visually indistinguishable after the LDA step, illustrating the power of separation-enhancing projections.